

Dynaωo Model-Validation Tool: initialization routines for synchronous generators in DTR Sheets

Grupo AIA

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Contents

Abstract

This document describes in detail the routines within the Dynaωo Model-Validation tool that calculate the initialization values for all elements in the various DTR Sheets [**DTRhome**].

1 Overview

Although the producer is expected to supply his own model for the generator, he cannot be asked to also provide the correct initialization values for all the the case scenarios prescribed by the DTR Sheets. Such initialization is dependent on many other parameters that vary from sheet to sheet, such as the connection reactances, the values of P, Q, or V at the connection point (PDR bus), etc. Therefore the Dynaωo Model-Validation tool performs its own internal calculations to find out the correct initialization values for the generator, auxiliary load, etc.; on a case-by-case basis.

For the moment being, *the approach consists in assuming that the topology on the producer's side is restricted to a small set of variants*, which enables us to calculate the quantities analytically by means of relatively simple load-flow (LF) formulas. In the future we do not rule out some other approach providing greater freedom, such as running the cases through one or more passes in *DynaFlow* [**Dynawo**] until a valid steady state is achieved.

2 Allowed topology variants

For our analytic LF approach to work, the topology on the producer's side *must* be one of these eight variants:

S: single gen/WT/PV unit

S+i: single gen/WT/PV unit + Line (represents an “internal network”)

S+Aux: single gen/WT/PV unit + Aux Load

S+Aux+i: single gen/WT/PV + Aux Load + Line

M: multiple WT/PV units

M+i: multiple WT/PV units + Line

M+Aux: multiple WT/PV units + Aux Load

M+Aux+i: multiple WT/PV units + Aux Load + Line

Note for Zone-1 tests: for the isolated tests of individual WT/PV units without plant-level control (i.e., the so-called “Zone 1” tests), the only allowed topology is “S”.

As we can observe, there are here two main groups: single-unit vs. multiple-unit topologies. Then in each case one may model, if desired, an “internal network” via a single Line object, and auxiliary loads via a single Load object. These topologies are depicted in Figs. ??–??. If the producer’s topology deviates from these assumptions, our analytic LF calculations will result in incorrect initialization parameters for the generators and therefore the simulations may start at an unstable state.

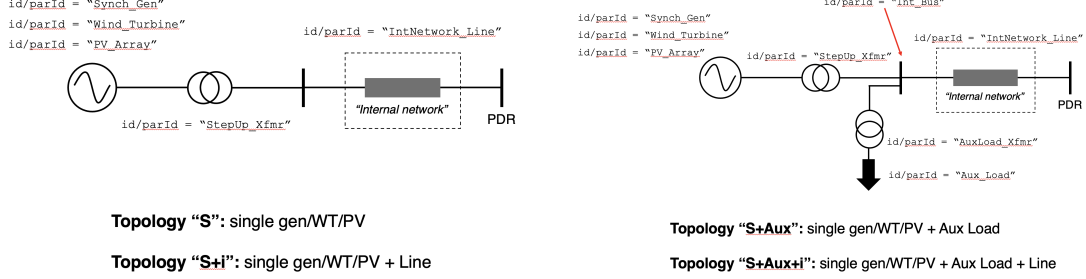


Figure 1: Allowed topologies for cases with a single unit.

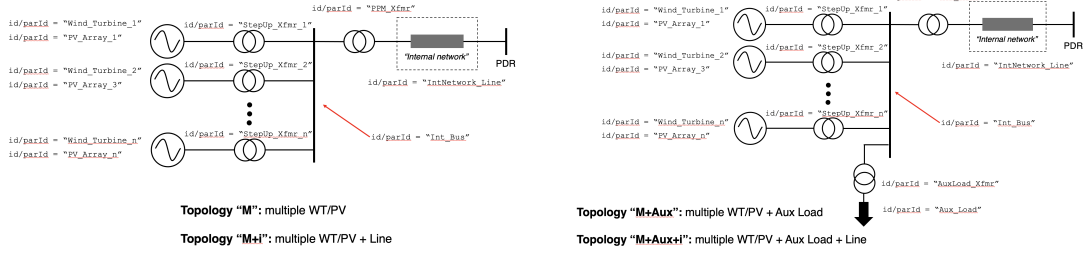


Figure 2: Allowed topologies for cases with multiple units.

3 Strategy for the calculation

Looking at the above topologies, and knowing that:

- the topology on RTE’s side always contains an Infinite Bus (except Fiche I10, in which there are only loads)
- the magnitudes V , P , Q are always prescribed at the PDR bus

It is easy to see that a general strategy for calculation is as follows:

1. First choose the angle reference (zero) to be at the PDR. Therefore all complex magnitudes V , I , S are known at the PDR bus.
2. Start by calculating the complex voltage at the Infinite Bus. This is the only initialization value we will need on RTE’s side. In addition, we may keep the value of the angle in order to perform a global shift at the end, in case we wanted the Infinite Bus to be the angle reference instead (although this is not really necessary).
3. We then proceed from the PDR bus towards the other side, section by section. The calculation will vary depending on the topology:
 - If there is only a line or a transformer followed by a generator unit, the calculation simply involves the formulas of a pi-model equivalent.

- If there is an auxiliary load, the calculation will involve a second-order algebraic equation (a standard two-bus load-flow), since this load specifies a constant PQ value.
- If there are multiple generating units, the calculation needs one additional input from the user: the sharing fractions of their respective PQ injections (we obtain this from the initialization values given by the user in his `Producer.par` file).

4 LF calculations for Topologies S and S+i

We start by noticing that the DTR Sheets specify three quantities at the point of connection, the PDR bus: voltage modulus and the complex flow $S = P + jQ$. This allows one to easily propagate the complex voltage and complex currents on either side of the PDR bus, *independently*. Actually, these LFs do not involve any quadratic equation at all, as it will be shown now (this will be different in the next sections, when a PQ auxiliary load is present).

The plan proceeds as follows, in outline mode:

- First LF: calculate the complex voltage at the RTE bus, needed to initialize the Infinite Bus model. We may use this voltage angle to perform a global phase shift, so that this bus becomes the reference (i.e., angle zero).
- Second LF (case S+i): calculate the complex voltage and current at the other end of the line representing the producer's "internal network", if it is modeled.
- Third LF: calculate the complex voltage and current on the low-voltage side of the generator transformer, thus obtaining the generator PQ output as well.

4.1 First LF: RTE's side

Let us focus first on the problem to the right of the PDR bus in Fig. ??, that is, RTE's network side. We start with this side first because, once V_∞ is solved for, we can shift phase globally so that V_∞ sets the reference (zero angle). This is not strictly necessary, but it is more convenient for analysts when they look at the results. Having in mind the pi-model equivalent of the connection line(s) and using current conservation plus the fundamental power equation $S = VI^*$, one has:

$$I_{\text{PDR}} = \frac{S_{\text{PDR}}^*}{V_{\text{PDR}}^*} - V_{\text{PDR}} Y_1^{(\text{sh})} = (V_{\text{PDR}} - V_\infty) Y^{(\text{tr})}$$

Since we will be repeating the same sort of calculation to the left of the PDR bus, it is useful to make this calculation more general by rewriting it in terms of a pi-model equivalent circuit as shown in Fig. ??:

$$I_{1 \rightarrow 2} = \frac{S_1^*}{V_1^*} - V_1 Y_1^{(\text{sh})} = (V_1 - V_2) Y^{(\text{tr})}$$

Rearranging this equation,

$$\frac{S_1^*}{V_1^* Y^{(\text{tr})}} - V_1 \frac{Y_1^{(\text{sh})}}{Y^{(\text{tr})}} = V_1 - V_2,$$

we easily obtain the complex voltage at terminal 2:

$$V_2 = V_1 \left(1 + \frac{Y_1^{(\text{sh})}}{Y^{(\text{tr})}} \right) - \frac{S_1^*}{V_1^* Y^{(\text{tr})}} \quad (1)$$

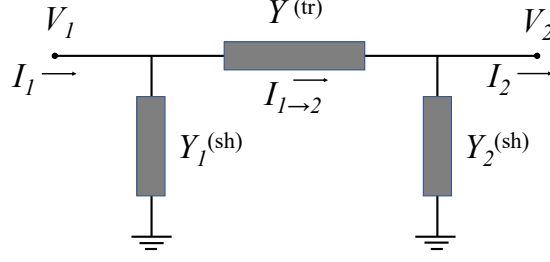


Figure 3: This is the pi-model circuit involved in the basic “loadflow” that we solve at several steps of the procedure. In our calculations, Terminal 1 always represents the bus where both complex voltage and complex current (or, equivalently, complex voltage and P & Q) are known. The complex voltage and current on terminal 2 are calculated through simple algebra (so this is not really a loadflow problem, strictly speaking). Notation: (tr) stands for *transmission* branch; (sh) stands for *shunt* admittance.

If instead of an infinite bus we had modeled a real generator on RTE’s side (as is the case of Fiche I8), we would also need to calculate the PQ injection of the generator. For that, we calculate the corresponding current I_2 . Using again the sign conventions indicated by the arrows in Fig. ??, this current is:

$$I_2 = (V_1 - V_2)Y^{(tr)} - V_2Y_2^{(sh)} \quad (2)$$

so that the total complex injection at the bus is:

$$S_2 \equiv P_2 + jQ_2 = V_2I_2^* \quad (3)$$

And finally the initialization values of P and Q for such generator are straightforward to obtain, simply by subtracting from S_2 the value (with the proper sign) of the equivalent PQ loads, if any are present (see Fiche I8).

4.2 Second LF: the producer’s “internal network” (case S+i)

If the producer has modeled an internal network (by means of a Line object), we will need to propagate the complex voltage and current to the other side, the internal bus.

The calculation is exactly the same as in the previous subsection, using again expressions (??) and (??). We will keep the convention that terminal 1 is the one at which magnitudes are known (in this case, the PDR bus), and terminal 2 is where we need to calculate voltages, currents, and so on. The signs of the resulting currents and power flows will be correct if we are consistent with the arrows: in this particular case here, it means that the values of inputs $P_1 + jQ_1$ need to have their signs flipped, compared to the values used in the previous subsection. To be sure, P_1 is positive in the previous subsection and it is negative here, as per the Dynawo convention (which is, “loads are positive”).

Therefore we will be using a common function `calc_pimodel_pf()`, but we will have to be careful when handling the signs of its inputs and outputs.

4.3 Third LF: the generating unit

As it can be seen in the diagram for topologies S and S+i in Fig. ??, it now suffices to calculate the voltage and current at the low-voltage side of the generator’s step-up transformer.

Again, the calculation is exactly the same as in the previous two subsections. This time terminal 1 is the HV side of the step-up transformer (which is the the PDR bus in case of topology S, and the internal bus in case of topology S+i). Therefore we will be

calling again the common function `calc_pimodel_pf()`, being careful when handling the signs of inputs and outputs. One thus obtains the complex voltage with expression (??) and the PQ output of the generator through expressions (??) and (??).

5 LF calculations for Topologies S+Aux, S+Aux+i

The calculations for these topologies are very similar to the previous ones. The only difference lies in the third LF, since the presence of an auxiliary load forces us to calculate a proper two-bus load-flow problem, i.e. solve a quadratic equation (for which it is straightforward to get the solution algebraically, in closed form).

The plan proceeds as follows, in outline mode:

- First LF: same as in the case of topologies S and S+i.
- Second LF (case S+Aux+i): same as in the case of topology S+i.
- Third LF: calculate voltage and current on the low-voltage side of the *auxiliary load* transformer first. We need to know this current in order to calculate the next and final step.
- Fourth LF: calculate voltage and current on the low-voltage side of the *generator* transformer, thus obtaining the generator PQ output as well.

5.1 Third LF: the auxiliary load circuit

As it can be seen in Fig. ?? (right hand side), the generator output PQ cannot be calculated until we know how much current is going into the auxiliary load circuit vs. the generator circuit.

Since the PQ consumption of the load is specified, and the complex voltage at the HV side of the load transformer is known, one can calculate the voltage at the load and thus the current. This is a standard two-bus power flow problem, which involves solving a quadratic equation.

Let us solve it in closed form. The circuit is again a simple pi-model model representing the auxiliary load transformer. As in the rest of this paper, we adopt the convention that terminal 1 represents the bus where both voltage and current are known—although in this particular case, the current is unknown! Terminal 2 is where the auxiliary load is connected to. Therefore equation we need to solve is:

$$(V_1 - V_2)Y^{(\text{tr})} = \frac{S_2^*}{V_2^*} + Y_2^{(\text{sh})}V_2,$$

where V_1 is known, S_2 is known (the load's PQ), and V_2 should be solved for. We will solve this in terms of the real and imaginary parts of V_2 . To prepare for that, let us rearrange this equation:

$$(Y^{(\text{tr})} + Y_2^{(\text{sh})})V_2 - Y^{(\text{tr})}V_1 + \frac{S_2^*}{V_2^*} = 0$$

and now divide the equation by suitable constants:

$$\left(\frac{Y^{(\text{tr})} + Y_2^{(\text{sh})}}{Y^{(\text{tr})}} \right) \frac{V_2}{V_1} - 1 + \left(\frac{S_2^*}{Y^{(\text{tr})}|V_1|^2} \right) \frac{V_1^*}{V_2^*} = 0$$

so that we can redefine the variable and the constants as follows:

$$\begin{aligned} v &\equiv \left(\frac{Y^{(\text{tr})} + Y_2^{(\text{sh})}}{Y^{(\text{tr})}} \right) \frac{V_2}{V_1} \\ \sigma &\equiv \frac{\left(Y^{(\text{tr})} + Y_2^{(\text{sh})} \right)^* S_2^*}{|Y^{(\text{tr})}|^2 |V_1|^2} \end{aligned} \tag{4}$$

in order to simplify the equation:

$$v - 1 + \frac{\sigma}{v^*} = 0 \quad (5)$$

It is now easy to solve this equation in the complex variable v by using the real and imaginary parts, $v_x + jv_y$. Let us first multiply by v^* :

$$|v|^2 - v^* + \sigma = 0.$$

The two equations are then:

$$\begin{aligned} v_x^2 + v_y^2 - v_x + \sigma_x &= 0 \\ v_y + \sigma_y &= 0 \end{aligned}$$

From this we readily obtain $v_y = -\sigma_y$, and therefore arrive at this quadratic equation for v_x :

$$v_x^2 - v_x + \sigma_x + \sigma_y^2 = 0$$

For which the solution is:

$$v_x = \frac{1}{2} \pm \sqrt{\frac{1}{4} - \sigma_x - \sigma_y^2}$$

Here the correct solution branch is obviously the one with the plus sign, as this would yield a high voltage ($v = 1$) if the loads were zero ($\sigma = 0$), as expected. Therefore the complex voltage is:

$$v = \frac{1}{2} + \sqrt{\frac{1}{4} - \sigma_x - \sigma_y^2} - j\sigma_y \quad (6)$$

from which one readily obtains V_2 by undoing the definitions in Eq. (??).

Lastly, the net current that flows from terminal 1 into the auxiliary loads circuit can be calculated as:

$$I_{\text{aux}} = (V_1 - V_2)Y^{(\text{tr})} + V_1Y_1^{(\text{sh})} \quad (7)$$

This quantity will be needed in the next section.

5.2 Fourth LF: the generator circuit

Here the calculation is very similar to the third LF in the case of topologies S and S+i, except that we first need to subtract from I_1 the current I_{aux} that is drawn by the auxiliary loads circuit (which was calculated in the previous subsection). Once we do that, both voltage and current are known at terminal 1 and one can calculate voltage and current at terminal 2 (i.e., the generator terminals, also the LV side of the step-up transformer).

Then the PQ output of the generator is simply given by $S_{\text{gen}} = S_2 = V_2I_2^*$.

6 LF calculations for Topologies M and M+i

The procedure is almost the same as that of Topologies S and S+i, with one crucial difference: when attempting to perform what we called the third LF, one needs more data; namely, how the net injection is *shared* among the several generating units. We obtain these proportionality factors for P and Q from the user input. More precisely, we read the existing initialization values of P_0 and Q_0 for each generator k in the `Producer.par` file provided by the user, and calculate the proportionality factors in a straightforward way:

$$\begin{aligned} \lambda_P^{(k)} &= \frac{P_0^{(k)}}{\sum_{i=1}^n P_0^{(i)}} \\ \lambda_Q^{(k)} &= \frac{Q_0^{(k)}}{\sum_{i=1}^n Q_0^{(i)}} \end{aligned} \quad (8)$$

where n is the number of generating units. We then simply apply these proportionality factors to the net injection S_1 of equation (??), to obtain the complex voltages and PQ values at each generator terminals, with equations (??)–(??).

7 LF calculations for Topologies M+Aux and M+Aux+i

The procedure follows the same steps as those for Topologies S+Aux and S+Aux+i, up to the third LF (for the auxiliary load). Then, the fourth loadflow is different: one first needs to subtract from I_1 the current I_{aux} that is drawn by the auxiliary load circuit. The new net value of the injection thus becomes $S_1 = V_1(I_1 - I_{\text{aux}})^*$, and we can then calculate the complex voltages and PQ values at each generator terminals as in the previous section, using their corresponding $\lambda_P^{(k)}, \lambda_Q^{(k)}$ sharing factors.

Appendices

A Pi-model equivalent parameters

In this section we describe how to obtain the pi-model parameters $Y^{(tr)}, Y_1^{(sh)}, Y_2^{(sh)}$ from the physical parameters that we can read from the Dynaωo case files.

A.1 Single line

This is quite simple, as Dynaωo’s Line model is already in the form of a pi-model equivalent. Pay attention, though: the shunt admittances are already divided by two (they follow Eurostag’s convention: they are “half-admittance”). Therefore one has:

$$\begin{aligned} Y^{(tr)} &= \frac{1}{\text{line_RPu} + j \text{line_XPu}} \\ Y_1^{(sh)} &= \text{line_GPu} + j \text{line_BPu} \\ Y_2^{(sh)} &= \text{line_GPu} + j \text{line_BPu} \end{aligned} \quad (9)$$

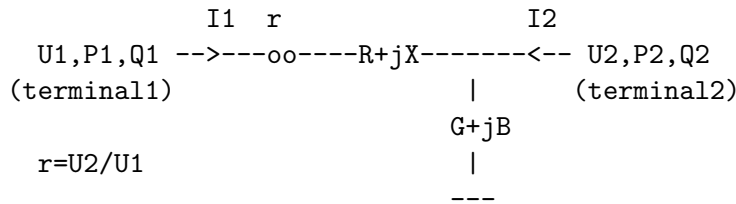
A.2 Multiple parallel lines

In this case one would simply *add* the admittances, as both the transmission and the shunt admittances are connected in parallel to each other, respectively.

$$\begin{aligned} Y^{(tr)} &= \sum_{i=1}^n \frac{1}{\text{line_RPu}_i + j \text{line_XPu}_i} \\ Y_1^{(sh)} &= \sum_{i=1}^n \text{line_GPu}_i + j \text{line_BPu}_i \\ Y_2^{(sh)} &= \sum_{i=1}^n \text{line_GPu}_i + j \text{line_BPu}_i \end{aligned} \quad (10)$$

A.3 Transformers

For transformers, one needs to pay attention to the internal modeling used in Dynaωo. This is the equivalent circuit and conventions:



In the following, we will use the notation $Z \equiv R + jX$ (transmission branch) and $Y_m \equiv G + jB$ (magnetizing branch). We will use this now this circuit to obtain the values of its pi-model equivalent $Y^{(tr)}, Y_1^{(sh)}, Y_2^{(sh)}$ in terms of Z and Y_m . Note that, for this calculation, we will be using the convention that current $I2$ flows in the other direction (see Fig. ??), but this does not affect the final formulas.

Let us first recall the pi-model circuit in Fig. ?. It is straightforward to write the circuit equations:

$$\begin{aligned} I_1 &= (V_1 - V_2)Y^{(tr)} + V_1Y_1^{(sh)} \\ I_2 &= (V_1 - V_2)Y^{(tr)} - V_2Y_2^{(sh)} \end{aligned}$$

We will re-arrange it in this form, which reminds us better of the familiar matrix equation $I = YV$:

$$\begin{aligned} I_1 &= \left(Y^{(\text{tr})} + Y_1^{(\text{sh})} \right) V_1 - Y^{(\text{tr})} V_2 \\ I_2 &= Y^{(\text{tr})} V_1 - \left(Y^{(\text{tr})} + Y_2^{(\text{sh})} \right) V_2 \end{aligned} \quad (11)$$

Now the aim is to obtain the analogous set of equations for the transformer's equivalent circuit, so that we can then identify each of these four coefficients between the two.

The transformer circuit can be solved from first principles as follows. We will use the following notation:

- e_1, i_1 : voltage and current at the ideal section of the transformer, on terminal 1 side.
- e_2, i_2 : voltage and current at the ideal section of the transformer, on terminal 2 side.

Looking at the diagram, we readily observe that $e_1 = V_1$ and $i_1 = I_1$. Now the two fundamental relations of the transformer are the following:

$$\begin{aligned} r &\equiv \frac{e_2}{e_1} && \text{definition of the tap ratio} \\ e_1 i_1^* &= e_2 i_2^* && \text{energy conservation (ideal transformer)} \end{aligned}$$

For I_1 , we obtain, step by step:

$$\begin{aligned} I_1 &= i_1 \\ &= \frac{e_2^*}{e_1^*} i_2 \\ &= r i_2 \\ &= \frac{r}{Z} (e_2 - V_2) \\ &= \frac{r}{Z} (r e_1 - V_2) \\ &= \frac{r^2}{Z} V_1 - \frac{r}{Z} V_2 \end{aligned}$$

And for I_2 :

$$\begin{aligned} I_2 &= i_2 - V_2 Y_m \\ &= \frac{1}{Z} (e_2 - V_2) - V_2 Y_m \\ &= \frac{1}{Z} (r V_1 - V_2) - V_2 Y_m \\ &= \frac{r}{Z} V_1 - \left(\frac{1}{Z} + Y_m \right) V_2 \end{aligned}$$

Equating now the coefficients in these two last equations to those of Eq.(??), one finally obtains:

$$\begin{aligned} Y^{(\text{tr})} &= \frac{r}{Z} \\ Y_1^{(\text{sh})} &= \frac{r(r-1)}{Z} \\ Y_2^{(\text{sh})} &= \frac{1-r}{Z} + Y_m \end{aligned} \quad (12)$$